# Divergence Based Non-Negative Matrix Factorization for top-N Recommendations

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# Overview


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5. Our Proposed Approach (TNMF)
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# Introduction

# What is a Recommendation Systems?

# Rating systems

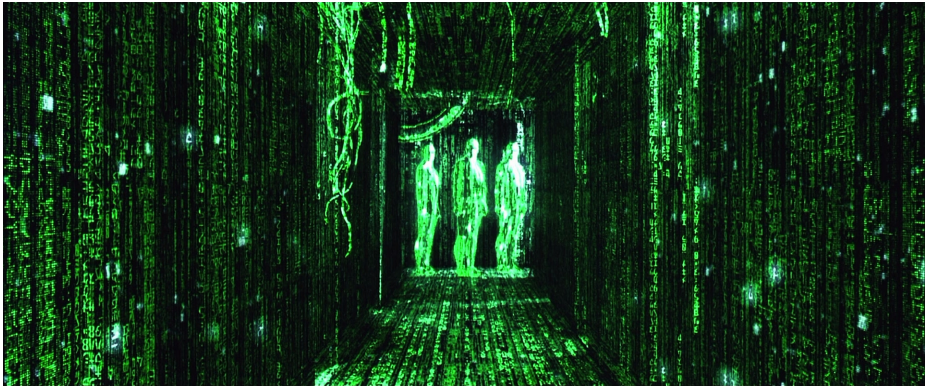
# Motivation

- Some form of distance function are mostly used to measure the difference between original and reconstructed matrices.
  - Inefficient to capture the continuous randomness
- To compute the differences between users rating distribution, we choose Kullback-Leibler divergence.
  - Information-based measure of disparity
  - Having convexity properties.

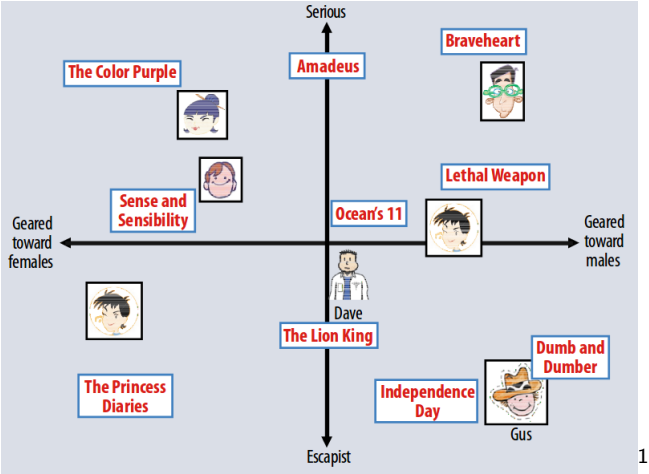
# Recommendation Systems

- Content-based systems.
- Collaborative filtering systems.
- Hybrid recommender systems.

# Matrix Factorization



# A latent space representation




[1]

[1] The picture is taken from Y. Koren et al. (2009). *Matrix Factorization Techniques for Recommender Systems.* Computer 42 (8)

# Known factorization models (1/5)

**Principal Component Analysis**(PCA)

- transform data to a new coordinate system.
- variances by any projection of the data lies on coordinates in decreasing order

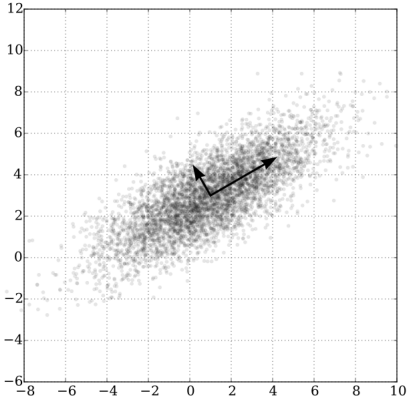
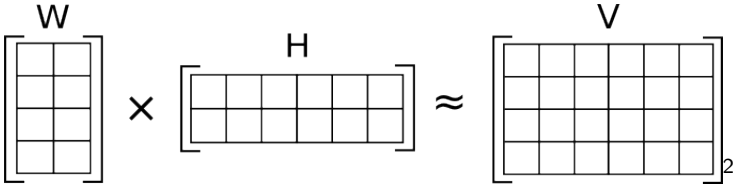



Figure: By Nicoguaro - Own work, CC BY 4.0, https://commons.wikimedia.org/

# Known factorization models (2/5)

**Matrix Factorization**(MF)




[2]By Qwertyus - Own work, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=29114677

# Known factorization models (3/5)

**Singular Value Decomposition**(SVD)

$$\Phi = W^{n\times k}\Sigma^{k\times k}H^{n\times k^\top}$$

- $W^\top W = I$, $H^\top H = I$.
- column vectors of W are orthonormal eigenvectors of $\Phi\Phi^\top$.
- column vectors of H are orthonormal eigenvectors of $\Phi\Phi^\top$.
- $\Sigma$ contains eigenvalues of $W$ in descending order.

PCA, SVD computed algebraically

- $\Phi$ is a **BIG** and **SPARSE** matrix.
- Approximations of PCA[3] and SVD[4].

[3] T.Raiko et al. (2007). Principal Component Analysis for Sparse High-Dimensional Data.Neural Information Processing, LNCS. 4984

[4] A.K. Menon and Ch. Elkan (2011). Fast Algorithms for Approximating the Singular Value Decomposition. ACM Trans. Knowl. Discov. Data 5 (2).

# Known factorization models (4/5)

## MF - rating prediction

Recommendation task

- to find $\hat{\Phi} : \mathcal{U}, \mathcal{I} \rightarrow \mathbb{R}$ such that the accuracy $acc(\hat{\Phi}, \Phi, \mathcal{T})$ is maximal.
- Training $\hat{\Phi}$ on $\mathcal{D}$ such that empirical loss $err(\hat{\Phi}, \Phi, \mathcal{D})$ is minimal.

A simple, approximative MF model

- $k$ is the number of latent factors.
- User component, $\mathcal{U} = X^{m \times k}$
- Item component, $\mathcal{I} = Y^{n \times k}$
- $\Phi^{m \times n} \approx \hat{\Phi}^{m \times n} = XY^{\top}$
- Predicted rating, $\hat{\Phi}_{ij} = x_i y_j^{\top}$

# Known factorization models (5/5)

## MF - rating prediction

The **loss** $err(\hat{\Phi}, \Phi, \mathcal{D})$ function

- square loss: $err(\hat{\Phi}, \Phi, \mathcal{D}) = \sum_{(i,j)\in\mathcal{D}}(\Phi_{ij} - \hat{\Phi}_{ij})^2$

the **objective** function

- **Regularization** term $\lambda \geq 0$ to prevent overfitting.
- Penalize the magnitude of the parameters.
- $F(\hat{\Phi}, \Phi, \mathcal{D}) = \sum_{(i,j)\in\mathcal{D}}(\Phi_{ij} - x_i y_j^\top)^2 + \lambda(||X||^2 + ||Y||^2)$

The **task** is to to find parameters $X$ and $Y$ such that, given $\lambda$, the objective function $F(\hat{\Phi}, \Phi, \mathcal{D})$ is minimal.

# Gradient Descent

*How to find a minimum of an "objective" function $F(\Theta)$?*

- In case of MF, $\Theta = XY^\top$, and
- $F(\Theta)$ refers to the error of approximation of $\Phi$ by $XY^\top$

<u>Gradient Descent</u>

**Input:** $F, \alpha, \Sigma^2$, *stopping criteria*

initialize $\Theta \sim \mathcal{N}(0, \Sigma^2)$

**repeat**

$\Theta \leftarrow \Theta - \alpha \frac{\delta F}{\delta \Theta}(\Theta)$

**unitl** approximate minimum is reached

**return** $\Theta$

<u>Stopping criteria</u>

- $\Theta^{old} - \Theta \leq \epsilon$
- maximum number of iterations reached
- a combination of both

# Stochastic Gradient Descent

When

$$F(\Theta) = \sum_{i=1}^{n} F(\Theta)$$

<u>Stochastic Gradient Descent</u>

**Input:** $F_i, \alpha, \Sigma^2, \textit{stopping criteria}$

initialize $\Theta \sim \mathcal{N}(0, \Sigma^2)$

**repeat**

  **for all** $i$ in random order **do**

    $\Theta \leftarrow \Theta - \alpha \frac{\delta F_i}{\delta \Theta}(\Theta)$

  **end for**

**unitl** approximate minimum is reached

**return** $\Theta$

# MF with Stochastic Gradient Descent

Updating parameters iteratively for each data point $\Phi_{ij}$ in the opposite direction of the gradient of the objective function at the given point until a convergence criterion is fulfilled.

# MF with Stochastic Gradient Descent - Example

Lets have the following hyper-parameters:
$K = 3, \alpha = 0.0002, \beta = 0.02, iter = 5000$

$\Phi =$

| | | | |
|---|---|---|---|
| 5 | 3 | 0 | 1 |
| 4 | 0 | 0 | 1 |
| 1 | 1 | 0 | 5 |
| 1 | 0 | 0 | 4 |
| 0 | 1 | 5 | 4 |

# MF with Stochastic Gradient Descent - Example

Results are:

$X =$

| | | |
|---|---|---|
| -0.04934113 | 1.34410185 | 1.77343084 |
| 0.03978801 | 1.18810803 | 1.3230008 |
| 2.01185337 | 0.51518384 | 0.49810045 |
| 1.58449972 | 0.64733736 | 0.27476875 |
| 1.54911873 | 0.41921957 | 1.0073149 |

$Y^\top =$

| | | | |
|---|---|---|---|
| 1.80019693 | 1.18141528 | 0.38356536 | -0.2664426 |
| -0.39075386 | 0.32691031 | 2.12877112 | 1.65417156 |
| 1.29524055 | 0.42334401 | 1.59878863 | 1.42570338 |

# MF with Stochastic Gradient Descent - Example

$\Phi =$

| | | | |
|---|---|---|---|
| 5 | 3 | 0 | 1 |
| 4 | 0 | 0 | 1 |
| 1 | 1 | 0 | 5 |
| 1 | 0 | 0 | 4 |
| 0 | 1 | 5 | 4 |

$\hat{\Phi} =$

| | | | |
|---|---|---|---|
| 4.99678094 | 2.93347608 | 4.44734609 | 0.99887879 |
| 3.96372815 | 2.38151041 | 3.67910781 | 0.99783584 |
| 1.0673425 | 0.84007158 | 5.02504173 | 4.96497519 |
| 0.96686521 | 0.8206766 | 4.0581564 | 3.97526862 |
| 1.95325792 | 1.19655369 | 4.91835471 | 4.02525881 |

# Our Proposed Approach

# What Matrix Factorization Assumes?

**Matrix factorization assumes that:**

- Each user can be described by $k$ attributes or features. For example, feature 1 might be a number that says how much each user likes sci-fi movies.
- Each item (movie) can be described by an analogous set of k attributes or features. To correspond to the above example, feature 1 for the movie might be a number that says how close the movie is to pure sci-fi.
- If we multiply each feature of the user by the corresponding feature of the movie and add everything together, this will be a good approximation for the rating the user would give that movie.

# Nonnegative Matrix Factorization

A typical NMF solves the following optimization problem:

$$\begin{aligned} \min_{X \in \mathbb{R}^{m \times r}, Y \in \mathbb{R}^{n \times r}} \quad & f(X, Y) \simeq \frac{1}{2} \| R - XY^\top \|_F^2 \\ \text{s.t.} \quad & X \geq 0, Y \geq 0 \end{aligned}$$

# Top-N Nonnegative Matrix Factorization (TNMF)

## Divergence based objective function

To compute the differences between users rating distribution Kullback-Leibler divergence is used.

- Information-based measure of disparity
- Having convexity properties.

$$\arg\min_{X,Y} \sum_{(i,j)\in\Upsilon} \mathcal{L}_{rank}\left(R_{ij}\log\frac{R_{ij}}{X^\top Y_{ij}} - R_{ij} + X^\top Y_{ij}\right) + \frac{\beta}{2}\left(\|X\|_F^2 + \|Y\|_F^2\right) \tag{1}$$

# Optimization Algorithm

## Gradients

$$X_i^{new} \leftarrow X_i^{old} - \eta \frac{\partial \mathcal{L}}{\partial X_i} \tag{2}$$

$$\frac{\partial \mathcal{L}}{\partial X_u} = \sum_{i=1}^{n} y_{ui} \frac{R_{ui}}{(X^\top Y)_{ui}} - \sum_{i=1}^{n} Y_{ui} + \beta X_u^\top \tag{3}$$

$$\frac{\partial \mathcal{L}}{\partial Y_v} = \sum_{i=1}^{m} x_{vi} \frac{R_{vi}}{(X^\top Y)_{vi}} - \sum_{i=1}^{m} X_{vi} + \beta Y_v^\top \tag{4}$$

# Top-N Nonnegative Matrix Factorization (TNMF)

## Update rules

$$X_u{}^{new} \leftarrow X_u{}^{old} - \eta \left( \sum_{i=1}^{n} y_{ui} \frac{R_{ui}}{(X^\top Y)_{ui}} + \sum_{i=1}^{n} Y_{ui} - \beta X_u^\top \right) \tag{5}$$

$$Y_v{}^{new} \leftarrow Y_v{}^{old} - \eta \left( \sum_{i=1}^{m} x_{vi} \frac{R_{vi}}{(X^\top Y)_{vi}} + \sum_{i=1}^{m} X_{vi} - \beta Y_v^\top \right) \tag{6}$$

# Projected Stochastic Gradient descent

**Input:** $\eta > 0, \beta > 0$, initialize $X_{k \times m} = \mathbf{0}$ and $Y_{k \times n} = \mathbf{0}$
**Set:** latent factors array, $k$
**repeat**

1. Randomly select $i, j$
2. Update $X_u$ by using (5)
3. Project the updated $X_u$ onto the feasible set:
   $X_u = \max(0, X_u)$
4. Update $Y_v$ by using (6)
5. Project the updated $Y_v$ onto the feasible set:
   $Y_v = \max(0, Y_v)$

**until** *converge*;
**return** $X, Y$

**Algorithm 1:** Projected Stochastic Gradient Descent

# Why not SVD?

## What is the issue with Singular Value Decomposition?

- Usually user-product matrix is very large. ($m, n$ are large numbers)
- SVD computation is too expensive.
- Complexity: $\mathcal{O}(mn \min(m, n))$

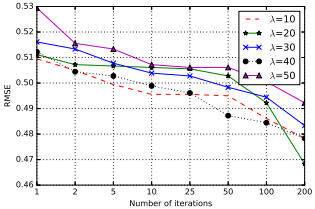
# Experimental Evaluations

# Dataset Statistics

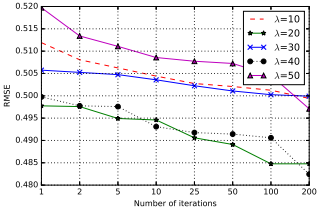
Table: Dataset statistics

| Sl. | Dataset | User | Item | Sparsity | Transact |
|---|---|---|---|---|---|
| 1 | *Yahoo!* music | 2689 | 994 | 3.95% | 6738 |
| 2 | *Yahoo!* movie | 2309 | 2380 | 0.18% | 10136 |
| 3 | ml-100k | 943 | 1682 | 6.30% | 100000 |
| 4 | Netflix | 83539 | 22 | 5.44% | 100000 |
| 5 | Book crossing | 2582 | 24009 | 0.12% | 12102 |
| 6 | Jester | 3000 | 100 | 71.01% | 213037 |

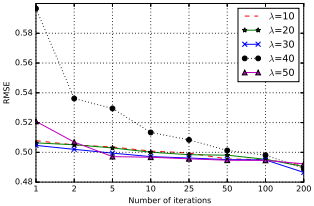
# Experimental Evaluation of TNMF



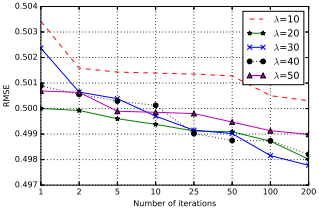

(a) Yahoo music



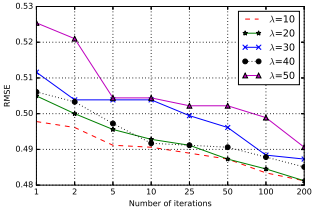

(b) Yahoo movies



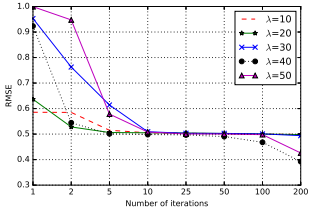

(c) ml-100k




(d) Netflix




(e) Book Crossing

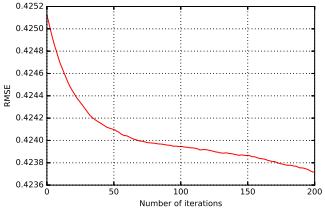



(f) Jester

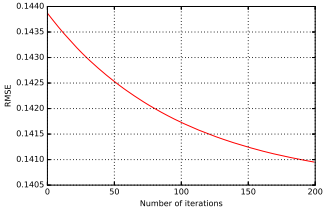
# Probabilistic Matrix Factorization

- Performs well on the large, sparse dataset.
- Extend PMF model to make interactive model capacity.
- According to their claim, designed method is nearly 7% better than Netflix.
- Use logistic function to bound the prediction range unlike simple linear Gaussian model.

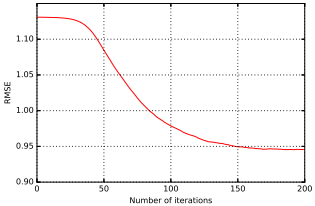
# Experimental Evaluation PMF



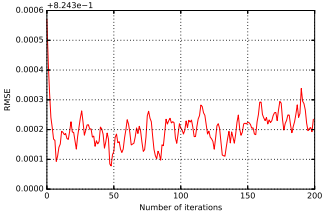

(a) Yahoo music



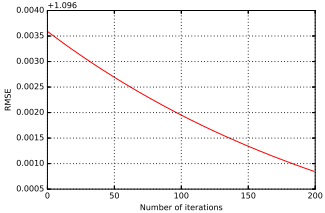

(b) Yahoo movies



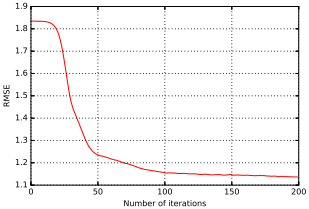

(c) ml-100k




(d) Netflix




(e) Book Crossing

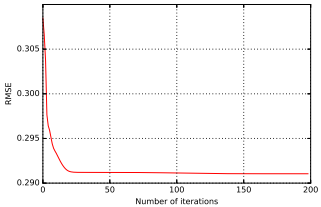



(f) Jester

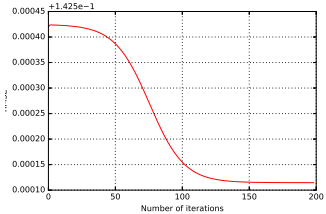
# Large-scale Parallel Collaborative Filtering for the Netflix Prize

- Problem: R contains many zeros or unknown ratings.
- Algorithm
  - Initialize item matrix M.
  - Fix M, solve user matrix U by minimizing objective function e.g., Loss function ($L^2$)
  - Fix U, solve M by minimizing similar function
  - Repeat steps until stopping criterion is satisfied
- Work with low-rank approximation of the user-item matrix.
- Performance of algorithm increases with number of features and number of ALS iterations

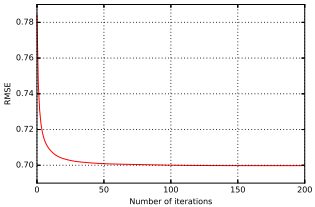
# Experimental Evaluation of ALS-WR



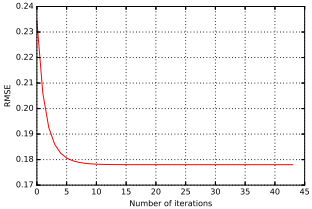

(a) Yahoo music



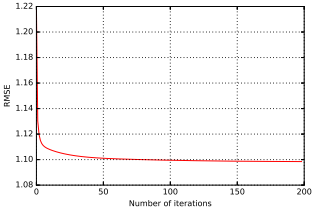

(b) Yahoo movies



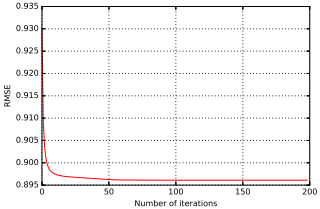

(c) ml-100k




(d) Netflix




(e) Book Crossing



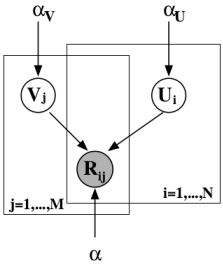

(f) Jester

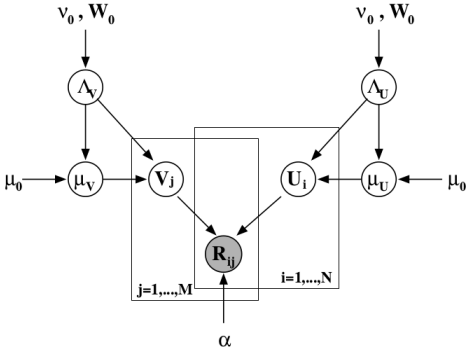
# Bayesian Probabilistic Matrix Factorization using Markov Chain Monte Carlo

- Model capacity is fitted automatically by integrating all parameters
- Efficiency improvement: By integrating Markov chain Monte Carlo Methods.
  - Result: higher prediction accuracy.

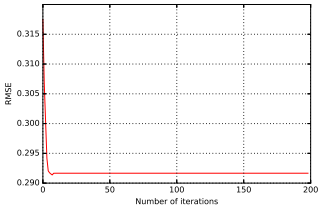
Challenges: Hard to determine when Markov chain is converged

# Bayesian Probabilistic Matrix Factorization using Markov Chain Monte Carlo

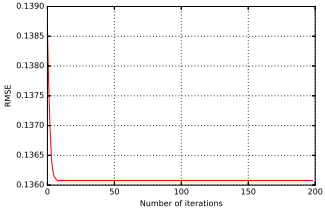


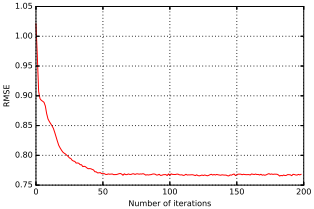
# Experimental Evaluation of BPMF (contd.)



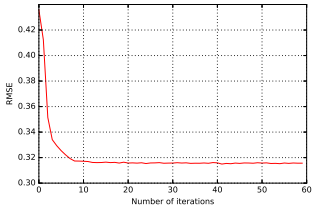

(a) Yahoo music



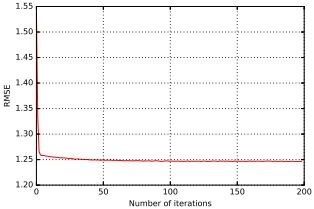

(b) Yahoo movies



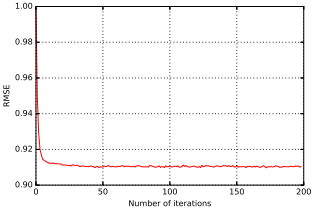

(c) ml-100k




(d) Netflix




(e) Book Crossing




(f) Jester

# Comparative Results

Table: RMSE with standard deviation for different datasets and methods.

| Methods | ml-100k | yahoo! music | yahoo! movies | Netflix | Jester | Book X |
|---|---|---|---|---|---|---|
| TNMF | **0.522±0.032** | 0.496±0.011 | 0.493±0.005 | 0.500±0.001 | **0.539±0.151** | **0.494±0.007** |
| PMF | 1.013±0.071 | 0.424±0.001 | 0.142±0.001 | 0.825±0.010 | 1.270±0.223 | 1.098±0.001 |
| ALS | 0.703±0.022 | **0.292±0.003** | 0.143±0.001 | **0.186±0.034** | 0.898±0.021 | 0.898±0.021 |
| BPMF | 0.782±0.040 | 0.299±0.011 | **0.137±0.002** | 0.323±0.025 | 0.913±0.023 | 0.913±0.023 |

# Conclusion

- Latent factors improve recommendation predictions.
- Regularized divergence based constrained non-convex minimization problem.
- Projected gradient descent to solve the problem.
- Improved hit rate and RMSE on most datasets compared to state-of-the-art methods.
- Limited applicability on sparse rating data.
- Pairwise ranking scheme can be included for improvement.

Questions?

https://github.com/enamul-haque/TNMF